

Q1) Given a joint distribution, assuming P_{ij} are positive

	$Y=1$	$Y=2$	$Y=3$
$X=1$	P_{11}	P_{12}	P_{13}
$X=2$	P_{21}	P_{22}	P_{23}

(a) Compute the marginal distribution $\text{Prob}(X=2)$ and $\text{Prob}(Y=1)$

$$\text{Pr}(X=2) = P_{21} + P_{22} + P_{23}$$

~~Sum~~ Sum across the row corresponding to $X=2$

Marginal of Y

$$\text{Pr}(Y=1) = P_{11} + P_{21} \quad (\text{Sum down the column for } Y=1)$$

(b) Compute $\text{Prob}(Y=1 | X=2)$

Using Conditional Probability

$$\text{Pr}(Y=1 | X=2) = \frac{\text{Pr}(X=2, Y=1)}{\text{Pr}(X=2)} = \frac{P_{21}}{P_{21} + P_{22} + P_{23}}$$

(c) Compute $\text{Prob}(X < Y)$

Event $X < Y$ means all (i, j) pairs where $i < j$

Possible Pairs:

$X=1, Y=2 \rightarrow$ Contributes P_{12}

$X=1, Y=3 \rightarrow$ Contributes P_{13}

$X=2, Y=3 \rightarrow$ Contributes P_{23}

$$\text{So } \text{Pr}(X < Y) = P_{12} + P_{13} + P_{23}$$

2) Given model

$$P(x|\theta; k) = \frac{1}{r(k)} \times \theta^k x^{k-1} \exp(-\theta x), \forall x \in (0, \infty)$$

k : no of people

x : observed waiting time

θ = unknown rate parameter we want to estimate

a) we assume the pairs (k_i, x_i) are i.i.d

likelihood for a single observation

$$P(x_i | k_i, \theta) = \frac{1}{r(k_i)} x_i^{k_i-1} \theta^{k_i} e^{-\theta x_i}$$

As the observations are independent

$$L(\theta) = \prod_{i=1}^n \frac{1}{r(k_i)} x_i^{k_i-1} \theta^{k_i} e^{-\theta x_i}$$

$$L(\theta) = \left(\prod_{i=1}^n \frac{1}{r(k_i)} x_i^{k_i-1} \right) \theta^{\sum_{i=1}^n k_i} \exp\left(-\theta \sum_{i=1}^n x_i\right)$$

b) Maximum Likelihood, we maximize with respect to θ

$$L(\theta) = \left(\sum_{i=1}^n k_i \right) \log \theta - \theta \sum_{i=1}^n x_i + \text{Constant}$$

$$\text{let } K = \sum_{i=1}^n k_i, \quad S = \sum_{i=1}^n x_i$$

Then

$$L(\theta) = K \log \theta - S \theta + \text{Constant.}$$

Differentiate w.r.t θ .

$$\frac{dL}{d\theta} = \frac{k}{\theta} - s$$

set derivative to 0

$$\frac{k}{\theta} - s = 0$$

$$\Rightarrow \frac{k}{\theta} = s \Rightarrow \theta = \frac{k}{s}$$

$$\theta = \frac{\sum_{i=1}^n k_i}{\sum_{i=1}^n x_i}$$

$$\Rightarrow \boxed{\frac{\sum k_i}{\sum x_i}}$$

3

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 & 0 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & 2 \end{bmatrix} \right)$$

Inverse of matrix is $\begin{bmatrix} 1/2 & 1/2 & 1/2 \\ 1/2 & 5/2 & 3/2 \\ 1/2 & 3/2 & 3/2 \end{bmatrix}$

inverse of $\begin{bmatrix} 5/2 & 3/2 \\ 3/2 & 3/2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -1 \\ -1 & 5/3 \end{bmatrix}$

Q(a) which two variables are independent with each other?

In Multivariate Gaussians, Two variables X_i and X_j are conditionally independent given all others j and only if $(\Sigma^{-1})_{ij} = 0$

$$(\Sigma^{-1})_{12} = 1/2 \rightarrow \text{non zero}$$

$$(\Sigma^{-1})_{13} = 1/2 \rightarrow \text{non zero}$$

$$(\Sigma^{-1})_{23} = 3/2 \rightarrow \text{non zero}$$

This means: No off-diagonal element is zero.

So, No two variables are independent.

(b) Given $Z = X_1 - aX_2 + bX_3$

We want to know if we can choose real numbers a, b so that $Z \perp X_3$

Since the variables are jointly Gaussian, independence $\Leftrightarrow$ zero covariance

$$\text{Cov}(Z, X_3) = 0$$

and because X_2 is already independent of X_1 (their covariance is 0) the conditional covariance is simply

$$\text{Cov}(Z, X_2 | X_1) = \text{Cov}(Z, X_2)$$

so we need to calculate $\text{Cov}(Z, X_2)$

$$\text{Cov}(Z, X_2) = \text{Cov}(X_1, X_2) - a \text{Var}(X_1) + b \text{Cov}(X_3, X_2)$$

using the covariance matrix

$$\text{Cov}(Z, X_2) = 0 - a - b$$

set equal to zero

$$-a - b = 0$$

$$b = -a$$

$$\text{Ex:1 } a = 1, b = -1$$

$$\text{Ex2 } a = 2, b = -2$$

$$\text{Ex } a = -3, b = 3$$

So, Yes it is possible, Any pair that satisfies $b = -a$ works

Q4 @ We have 1-D data points: 4, 10, 16, 20, 26

we need $k=3$ clusters with initial centroids

$$\mu_1 = 2, \mu_2 = 3, \mu_3 = 4$$

we will run k-means until convergence.

Iteration 0 (initial centroids) $\mu_1 = 2, \mu_2 = 3, \mu_3 = 4$

Iteration 1 Every point is closest to 4, so all points go to cluster 3: update only $\mu_3 = \frac{4+10+16+20+26}{5} = 15.2$

Iteration 2

4 and 10 are closer to 3 $\rightarrow$ go to cluster 2

16, 20, 26 go to cluster 3

$$\text{update } \mu_2 = \frac{4+10}{2} = 7, \mu_3 = \frac{16+20+26}{3} = 20.67$$

$$\text{Cov}(Z, X_3) = \text{Cov}(X_1, X_3) - a \{ \text{Cov}(X_2, X_3) + b \text{Var}(X_3) \}$$

$$\text{So. } \text{Cov}(X_1, X_3) = -1$$

$$\text{Cov}(X_2, X_3) = -1$$

$$\text{Var}(X_3) = 2$$

$$\begin{bmatrix} 3 & 0 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

$$\text{Cov}(Z, X_3) = (-1) - a(-1) + b(2) \Rightarrow -1 + a + 2b$$

$$-1 + a + 2b = 0$$

$$a + 2b = 1$$

Ex: Sol 1
if $b=0 \Rightarrow a=1$ so, $(a, b) = (1, 0)$

Ex: Sol 2

$$\text{if } b = \frac{1}{2} \Rightarrow a = 0 \quad \text{so } (a, b) = (0, \frac{1}{2})$$

Ex Sol 3

$$\text{if } b = -1 \quad a - 2 = 1 \Rightarrow a = 3 \quad (a, b) = (3, -1)$$

All are valid. So, Yes it is possible. The condition for independence is $a + 2b = 1$

(c) $Z = X_1 - aX_2 + bX_3$; we want to know whether it is possible to choose constants a, b such that

$$Z \perp X_2 \mid X_1 = x_1 \text{ for any fixed value } x_1 \in \mathbb{R}.$$

For Gaussian variables, it means The Conditional covariance between

Z and X_2 must be zero

5 a 0.0

b 0.0

c 0.0

d 0.0

e 0.0

f 1.0

g 0.0

h 1.0

i 1.0